

RMO(INDIA) – 1995

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. In triangle ABC, K and L are points on the sides BC (K being closer to B than L) such that $BC \cdot KL = BK \cdot CL$ and AL bisect $\angle KAC$. Show that AL is perpendicular to BK.
 2. Call a positive integer n good, if there are n integers, positive or negative, and not necessarily distinct, such that their sum and product are both equal to n. Show that integers of the form $4k+1 (k \geq 0)$ and $4l (l \geq 2)$ are good.
 3. Prove that among any 18 consecutive three digit numbers there is at least one number which is divisible by the sum of its digits.
 4. Show that the quadratic equation $x^2 + 7x - 14(q^2 + 1) = 0$, where q is an integer, has no integer root.
 5. Show that for any triangle ABC, the following inequality is true:
 $a^2 + b^2 + c^2 > \sqrt{3} \max \{ |a^2 - b^2|, |b^2 - c^2|, |c^2 - a^2| \}$, where a, b, c are the sides of the triangle.
 6. Suppose $A_1 A_2 \dots A_{21}$ is a 210 sided regular polygon inscribed in a circle with centre O. How many triangles $A_i A_j A_k, 1 \leq i < j < k \leq 21$, contain the point O as interior point.
 7. Show that for any real number x: $x^2 \sin x + x \cos x + x^2 + \frac{1}{2} > 0$.

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